Problem E
Mergers & Acquisitions

Time limit: 1 second

There are N corporations arranged in a line in Silicon Valley, numbered from 1 to N from left to right. Corporation i has a bank balance of P_i billion.

Because humans are inherently greedy, the CEO of a corporation may take over the corporation immediately to its left if and only if its bank balance is strictly greater than that corporation's balance.

Suppose corporation B takes over corporation A. Corporation A is permanently dissolved. Corporation B takes over A's territory, bringing B adjacent to whichever corporation was immediately to the left of A.

Corporation B's bank balance stays unchanged: it does not receive A's balance. It can then take over its new left neighbour if its balance is strictly greater. Corporations with equal balances cannot take over each other.

You are given Q independent queries. Each query contains two integers L and R. For that query, consider only the original corporations from position L to position R, inclusive. Corporations outside this segment do not participate.

The corporations perform takeovers until no valid takeover remains. If several takeovers are possible, they may be performed in any order.

For each query, determine how many corporations remain. Every query starts from the original arrangement. Takeovers in one query do not affect any other query.

Input

The first line contains two integers N and Q ($1 \le N, Q \le 2 \cdot 10^5$), the number of corporations and the number of queries.

The second line contains N integers $P_1, P_2, \ldots, P_N$ ($1 \le P_i \le 10^9$), the corporations' bank balances.

Each of the next Q lines contains two integers L and R ($1 \le L \le R \le N$), describing a query.

Output

For each query, print the number of remaining corporations on a separate line, in the order the queries appear in the input.

Sample Input 1	Sample Output 1
5 4 5 2 3 1 4 1 5 2 5 3 5 1 3	2 1 1 2